



OPEN Unmanned aerial vehicle routing based on frog-leaping optimization algorithm

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Routing in Unmanned Aerial Vehicle (UAV) networks is critical for effective data transfer and overall network performance. However, current UAV routing algorithms exhibit high latency, poor route selection, excessive energy consumption, and limited flexibility in changing network topologies. To overcome these limitations, this paper proposes a new routing strategy that uses the Shuffled Frog Leaping Algorithm (SFLA) to improve UAV network routing. Using a two-phase optimization approach considering Quality of Service (QoS), our system combines global exploration with local exploitation, unlike previous techniques. This hybrid method enables UAVs to dynamically change their trajectories, helping to choose the best path even in fast-changing surroundings. Our approach's self-adaptive population-based search mechanism accelerates convergence and removes a common weakness in traditional metaheuristic algorithms—premature standstill elimination—which determines its effectiveness. By constantly adjusting UAV routing patterns depending on energy economy, latency, and throughput characteristics, SFLA guarantees that UAV networks transmit effectively and consistently. Based on experimental data, our method outperforms benchmark alternatives in terms of energy use by 3.11%, latency by 5.14%, and network lifetime by 2.25%. These developments make our approach ideal for real-time applications including aerial surveillance and disaster response that call for high data transfer speeds and great energy economy.

Keywords Unmanned aerial vehicle (UAV), Routing algorithm, Frog-leaping algorithm, Aerial vehicles

From structural inspections and hazard mapping to monitoring oil spills and carrying out surveillance, Unmanned Aerial Vehicles (UAVs) have grown to be indispensable instruments in many disciplines¹. In disaster management, where they enable quick data collecting and support efficient planning for humanitarian relief, their adaptability is particularly important². Despite their great potential, using UAVs presents several difficulties³. Effective UAV networks manage dynamic topologies, high mobility, and low latency while also including seamless inter-drone communication, onboard data processing, and storage⁴. Particularly considering restrictions like limited flight times, energy consumption, and complicated multi-zone surveillance demands, the need for effective routing becomes ever more important as these networks grow⁵.

Creative solutions are necessary to address these difficulties; the Frog-Leaping Optimization Algorithm presents a bright future⁶. This method efficiently blends local and global search methods to offer a compromise between adaptability and accuracy⁷. Its speed and accuracy make it perfect for managing the dynamic and erratic character of UAV networks, guaranteeing quick and effective data delivery⁸. Moreover, its adaptability enables it to meet the several needs of various UAV applications, whether they entail high mobility, latency-sensitive activities, or complicated routing scenarios with several paths^{9,10}.

Current solutions sometimes suffer from high latency, ineffective routing, and too demanding computational requirements even with improvements in UAV routing algorithms. These challenges become especially more obvious in dynamic networks where UAVs typically move around and it is challenging to establish reliable communication paths. We introduce in this work a fresh routing method designed to surpass these constraints. Including Quality of Service (QoS) criteria and the Frog-Leaping Optimization Algorithm guarantees a more adaptable and efficient routing mechanism. To maximize data flow and general performance while yet increasing the speed, energy efficiency, and lifespan of UAV networks, we thus present a new routing technique using the Shuffled Frog Leaping Algorithm (SFLA). Combining the SFLA with QoS settings results in a dynamic, flexible, and energy-efficient UAV routing system which is the main originality of this work. Whereas conventional

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methods including Ant Colony Optimization (ACO), Energy-Efficient Neuro-Fuzzy Clustering (EENFC-MRP), and Particle Swarm Optimization (PSOA) have high computational complexity, ineffective route selection, or excessive energy consumption, our method uses a two-phase optimization strategy combining global exploration and local exploitation. Since it allows faster convergence, real-time route adaptation, and less computational costs, the technology fits large-scale UAV networks. Moreover, our system preserves low latency and best energy management utilizing a self-adaptive memplex-based architecture, so increasing UAV network scalability and endurance. These advances make SFLA-based UAV routing a reasonable and efficient solution for real-time applications including extensive aerial communication networks, surveillance, and disaster response. We have made the following contributions:

- Proposing a dynamic routing method that minimizes latency and improves efficiency;
- Incorporating QoS parameters to enhance the reliability of route selection;
- Combining local and global search techniques to optimize route discovery and maintenance;
- Improving routing adaptability to accommodate the dynamic topology of UAV networks;
- Enhancing overall system performance through reduced computational complexity.

The rest of this paper is organized as follows: Section “[Related work](#)” reviews similar studies, pointing out recent advancements and their drawbacks. Section “[UAV routing method based on frog leaping algorithm](#)” talks about a new way to direct traffic using SFLA. It explains the math involved, how it makes processes better, and how it interacts with drone networks. Section “[Evaluation of the results](#)” describes how we set up our experiments. It includes details about the simulation setup, key factors, and how we assess performance. We compare our approach to other UAV routing methods, discussing how complex and efficient our algorithm is, along with a detailed comparison to other advanced methods. Section “[Conclusion and future works](#)” concludes the paper by summarizing the key findings, highlighting its contributions, and suggesting ideas for future research.

Related work

Routing in UAV networks is a novel and rapidly growing research area with a limited set of research findings. Our research provides a comprehensive and comparative study of routing protocols for UAV networks. Routing protocols are classified based on network structure and protocol operation. Their main operations and distinctive features are summarized and compared in terms of various primary parameters such as routing policies and metrics, mobility patterns, data transmission techniques, energy efficiency, localization, and end-to-end delay. For different applications, the comparison results can help engineers and researchers select the most suitable protocol for their intended applications. In this regard, Aadil, et al.¹¹ addressed the short flight time and inefficient routing problems by using performance clustering. They first adjust the transmission power of UAVs by predicting their operational needs. The optimal transmission range has the minimum Packet Loss Rate (PLR) and better link quality, ultimately saving energy consumption during communication. Second, they use a variant of the K-Means Density clustering algorithm to select cluster heads. Optimal cluster heads increase cluster lifetime and reduce routing overhead. Also, Pustokhina, et al.¹² developed an energy-efficient fuzzy neural cluster-based topology structure with an EENFC-MRP meta-heuristic path-planning algorithm for UAVs. Their proposed model included the EENFC. The cluster-based routing processed and MRP model of EENFC utilized three input parameters: the remaining energy of the UAV, the average distance to the nearest UAVs, and the UAV’s degree for cluster construction. Additionally, Quantum Ant Lion Optimization (QALO)-based MRP is applied to select an optimal set of paths for inter-cluster UAV communication. Moreover, Qi, et al.¹³ proposed a clustering architecture, FASNET, with SDN cluster controllers and a collaborative controller to achieve hierarchical management and unified forwarding. Based on the designed architecture, they also proposed a distinct Traffic-Diversified Routing (TDR) protocol that runs in each cluster, aiming to guarantee specific QoS requirements for delay-sensitive and reliability-sensitive services. Different weights are assigned to different flows according to their delay sensitivity and importance levels. Specifically, a transmission reliability prediction model is introduced to TDR, considering both link availability and node transmission capability. Inspired by the optimization behavior of Glowworms Swarm Optimization (GSO) for cluster generation and management, Khan, et al.¹⁴ also proposed a self-organizing clustering scheme. Along with the Luciferin value and remaining energy of UAVs, cluster head selection and formation are grounded on connectivity with the ground control station. The mechanism for cluster management updated the Luciferin value depending on UAV positions using GSO behavior. For effective communication, they also suggested a path-choosing mechanism based on neighborhood range, remaining energy, and UAV position. Furthermore, Liu, et al.¹⁵ developed the Lightweight Trustworthy Message Exchange (LTME) system, which effectively combines trust management and encryption technologies for drone networks. Under the LTME system, a primary Ground Control Station (GCS) securely shares critical information with registered drones and routinely checks and updates their trust levels. Only trustworthy receiving UAVs could decode encrypted messages each reliable broadcasting UAV produced using the secret values it acquired. Every reliable receiving UAV precisely assessed whether the received messages and matching broadcasting UAVs were lightweight and reliable. They carried out thorough research and tests for both sLTME, a simpler variant of LTME, and another method. While requiring less computing and communication effort, the results revealed that their approaches were quite dependable and had many practical features. In several respects, they also performed far better than the present techniques. Besides, Guo, et al.¹⁶ propose a method called Intelligent Clustering Routing Approach (ICRA) to make routing more effective in UAV ad hoc networks (UANETs). ICRA consisted of three components: the routing module, the module for modifying clustering techniques, and the clustering module. Every node evaluated its usefulness during clusterings. Their system extends its lifetime in several environments by improving its grouping process using reinforcement learning, preserving a stable network. Their method taught how to evaluate, under different

scenarios, the value of particular points in a network. Based on the state of the current network, their clustering technique modified module used what it knows to find the optimal way to group objects. Utilizing special nodes to enable message delivery between groups, their approach lowered delays and enhanced the success of packet delivery in the routing phase. Many tests revealed that their model beats other approaches and performs rather well. Their approach outperformed the best current methods in clustering speed, stability, energy use, and service quality, according to the findings. Also, Using drones in vehicle-restricted zones and no-fly zones, Wei, et al.¹⁷ solved the time-dependent vehicle routing challenge. Their optimization model took time-dependent road networks, no-fly zones and vehicle-restricted zones into account on delivery paths. Minimizing the total cost—including vehicle dispatch costs, energy consumption costs for vehicles and drones, and time-window penalty costs—was their goal. Gurobi helped their model to be verified for accuracy. Their method begins with logistic mapping and reverse learning creating the starting population. It then optimized the starting population by enhancing the genetic operators and variable neighborhood search operators. Finally, Meskar and Ahmadi-Javid¹⁸ investigated how a user of such a shared delivery system might best choose the necessary bases among the current bases to execute her delivery activities. Thus, a location-routing problem is investigated where the drone launching centers and their paths to deliver goods define themselves. Energy consumption in all flight phases including takeoff, level flight, hovering, and landing is computed using a realistic energy function that combines load weight. Although the energy consumption is nonlinear, the problem is formulated as a mixed-integer linear programming model and strengthened by valid inequalities and a pre-processing algorithm allowing us to solve instances with up to 100 customers using off-the-shelf optimization solvers. Furthermore, presented is a heuristic approach to handle cases including up to 200 clients. Results showed that using the load-dependent energy formula is more important than imposing rigid flight duration restrictions for drones.

Although routing protocols for UAV networks have come a long way, reaching ideal performance across important criteria including energy efficiency, latency, throughput, and network lifetime is still difficult. Though creative, many current solutions depend on meta-heuristic algorithms or sophisticated clustering methods. These methods demand exact parameter tuning and can bring computational overhead, which might not always be feasible. Moreover, most current techniques deal with individual performance measures instead of providing a comprehensive enhancement in several QoS levels. This makes a more simplified and all-encompassing approach that strikes efficiency and computational feasibility as top priorities desperately needed. The proposed work offers a fresh routing strategy based on the frog-leaping optimization technique to close this difference. Among other QoS criteria, this method is especially meant to improve routing procedures by lowering energy consumption, hence minimizing latency, and so strengthening network lifetime and throughput. Simulation results show that this approach beats current solutions, so proving its efficiency in overcoming present obstacles and providing a stronger and more effective solution for UAV network routing.

UAV routing method based on frog leaping algorithm

This part addresses routing in UAVs using the frog-leap optimization technique. Providing paths with the best quality of service inside the network structure will help to implement the routing process in UAV network architectures. After that, the whole structure of the frog-leap optimization method is fully explained, and subsequently, the application of the frog-leap optimization method model is discussed to execute the routing process. Nowadays, drones are indispensable for many purposes including military observation, environmental monitoring, and disaster response. Running drone networks presents a major difficulty in developing efficient routing systems that enable rapid data transfer while having energy-saving capability. Conventional routing techniques cannot adapt to new network layouts and limited resources, hence they sometimes struggle in changing circumstances. This paper investigates a real-time, network of drones used for data collecting and transmission within a given area. Because they keep moving, the UAVs build a Mobile Ad-hoc Network (MANET) that alters form. This mobility results in frequent changes in network connections, which calls for fast adjusting routing mechanisms to maintain the dependability of communication. Using wireless connections, UAVs converse with ground control and one another. Changing signal conditions, interference, and distance all can influence these links. Environmental barriers, weather, drone positions, and other factors can affect the quality of communication, so altering data speeds and maybe create communication interruptions. Usually lacking a lot of power, each UAV runs on a limited energy source—typically batteries. Things like how much data is being sent, the jobs being handled, and aircraft movement influences energy use. Extended network performance depends on good energy management, which also affects. While determining the best course of action, routing decisions should help to lower energy consumption. To meet the Quality of Service (QoS) requirements of the application, the routing protocol must offer fast data transfer and quick reactions. Delivering data on time, minimizing any data loss, and ensuring sufficient internet speed for critical communications thus become priorities. The system should rapidly change to fit various network environments to preserve significant quality criteria. For uses including real-time monitoring and emergency services, this is vital. Our work intends to develop a routing system addressing the difficulties of UAV networks so guaranteeing efficient, reliable, and energy-saving communication.

Our system model consists of a multi-UAV network implemented over a designated area for real-time data transmission whereby UAVs form a dynamic ad hoc network acting as mobile nodes transmitting data to other UAVs or Ground Control Stations (GCS). The network topology is rather dynamic since UAVs migrate depending on predefined mobility patterns, which induces regular topological changes. Path loss, interference, and signal fluctuations affect wireless links—used by UAVs. Given UAV's low battery capacity, energy-efficient routing methods are essential to extend network lifetime and maintain high-quality service. The routing system must reduce latency, maximize throughput, and lower energy consumption if it is to guarantee constant data flow. To meet these challenges, we propose an SFLA-based routing framework that dynamically optimizes UAV routing paths depending on real-time network conditions and QoS constraints. Several constraints define

various limits for conventional UAV routing protocols resulting from high latency resulting from poor path selection and network congestion, too high energy consumption resulting from ineffective route planning, and limited scalability resulting from stationary clustering or centralized control mechanisms failing to adapt to large-scale UAV networks. Thus scalable for large UAV networks our approach solves these problems utilizing a metaheuristic optimization-based routing mechanism that dynamically chooses optimal paths, guarantees low latency and congestion-free routing, balances energy consumption by including UAV battery levels and transmission costs, and adapts to real-time topological changes. Combining QoS-aware routing with SFLA-based optimization offers our method a computationally efficient, adaptive, and scalable solution for UAV network communication.

Frog leaping algorithm

Memetic meta-heuristic algorithms include the SFLA. This method uses a memetic local search within frog subgroups. Information is shared in the SFLA not only during local but also global searches, so combining both search approaches. The SFLA is rather easy to use and shows great capacity for worldwide search. SFLA can solve many kinds of nonlinear, non-convex, and multimodal problems.

Steps of frog leaping algorithm

The meta-heuristic strategy of the frog leaping algorithm in UAVs is summarized in two main steps: global exploration and local exploration.

Global exploration: This step generally comprises the following steps:

Step 1: Initialization

Pick any values for M and N , where M represents the number of memeplexes and N is the number of UAVs in each one. Then, to find the total number of UAVs in the swarm, simply multiply them together: $f = M * N$.

Step 2: Virtual Population Generation

A sample of F virtual UAVs, denoted as $U(1), U(2), \dots, U(F)$, is drawn from the feasible space. The fitness value, $f(i)$, of each $U(i)$ is calculated by the $U(i) = (U_i^1, U_i^2, \dots, U_i^d)$ where d represents the number of decision variables.

Step 3: Ranking and Sorting of UAVs

The UAVs are sorted in descending order based on their fitness values and stored in the array $X = \{U(i), f(i), i = 1 \dots, F\}$. The position of the best UAV, P_x , is recorded among all UAVs, where $U.P_x = 1$.

Step 4: Dividing UAVs into Memeplexes

The array X is divided into M memeplexes (m, Y^M) , where each contains N UAVs¹⁹.

Step 5: Memeplex Evolution

This step involves the evolution of each memeplex. Each memeplex undergoes local search (using the frog leaping algorithm), as described below.

Step 6: Memeplex Combination

After a certain number of memeplex evolutions, the memeplexes are combined into X , such that the relationship $\{m, \dots, 1 = k, Y^k\} = X$ holds. The position of the best UAV in the population (P_x) is then updated.

Step 7: Convergence Check

In this step, the convergence criterion is checked. If the convergence condition is met, the process terminates. Otherwise, the process returns to Step 4 of the global search.

Local exploration step

In the fifth step of the global search, each memeplex evolves independently N times. After the memeplexes evolve, the algorithm returns to the global search for combination. The details of the local search within each memeplex are explained below:

Step 1: Initialization

L_n and I_m are set to zero, where I_m counts the number of memeplexes and L_n counts the number of evolution steps.

Step 2: $I_m = I_m + 1$

Step 3: $L_n = L_n + 1$

Step 2: Submemeplex Creation

This step involves creating a submemeplex. The goal is for the UAVs to move towards optimal positions by improving their memes. The method for selecting the submemeplex involves assigning higher weights to UAVs with higher performance and lower weights to UAVs with lower performance values. Weights are assigned using a triangular probability distribution, defined by $P_i = (1 + n) / (j - 1 + n)^2$. To construct the array *submemeplex* (Z), q UAVs are randomly selected from each memeplex of n UAVs. The UAVs in the submemeplex are sorted in descending order based on their fitness. The positions of the best and worst UAVs in the submemeplex are denoted by PB and PW , respectively.

Step 3: Worst UAV Position Correction

In this step, the position of the worst UAV in the *submemeplex* (the UAV with the lowest fitness value) is corrected. The new position of the worst UAV, $U(q)$, is calculated by $S + PW = U(q)$, where S is the step size (jump amount) of the UAV. If the new position is better than the previous one, then the new $U(q)$ replaces the old $U(q)$, and the process moves to step 8 of the local search. Otherwise, it proceeds to step 6 of the local search.

Step 4: Calculation of Step Size by PX

If a better result was not produced in step 5, the step size of the UAV is calculated and the new position, $U(q)$, is calculated by $S + PW$. If $U(q)$ is within the feasible space, the new fitness value, $f(q)$, is calculated. If the new $f(q)$ is better than the previous one, the new $U(q)$ replaces the old $U(q)$, and the process moves to step 8 of the local search. Otherwise, it proceeds to step 7 of the local search.

Step 5: Random Solution Generation

If the new position is not feasible or does not result in a better fitness value, a new UAV (r) is randomly generated within the feasible region and replaces the UAV (q) whose new position was not suitable for improvement. The fitness of the new UAV, $f(r)$, is calculated, and $U(q)$ is set equal to r , and $f(q)$ is set equal to $f(r)$.

Step 6: Memeplex Update

After modifying the meme of the worst UAV in the *submemeplex*, the frogs in Z are placed back in their original positions in Y_{im} . Y_{im} is then sorted in descending order based on the fitness value.

Step 7: Iteration Check

If $N > iN$, the process goes back to step 3 of the local search.

Step 8: Iteration Check

If $m > im$, the process goes back to step 1 of the local search. Otherwise, the process returns to the global search for the memeplex combination. The proposed method employs a two-step search strategy, local and global, to identify the optimal solution (ch). In this algorithm, UAVs constitute a population, with each UAV possessing a structure analogous to a chromosome in a genetic algorithm. The entire UAV population is partitioned into smaller groups, where each group represents a distinct type of UAV (ch) dispersed across various regions of the solution space. Subsequently, each group of UAVs initiates a localized, intensive search within its vicinity. Each UAV within a group is influenced by both its group members and other groups. After several iterations, a mixing process occurs, disseminating information among all groups to ensure convergence and the attainment of the optimal solution. The proposed algorithm employs a two-phase search strategy, global and local, to identify the best solution. The overall cost function for the proposed method is as follows Eqs. (1) and (2).

$$Di = rand() (Xb - Xw) \quad (1)$$

$$Xw_{new} = Xw_{old} + Di(-Dmax \leq Di \leq Dmax) \quad (2)$$

Formula (1) and (2) in the frog-leaping algorithm dynamically update UAV positions. Xb is the best-performing UAV (highest fitness), Xw is the worst-performing UAV (lowest fitness), and $rand$ is a random number in (0,1) introducing movement diversity. The step size, $Dstep$, is calculated as Eq. (3).

$$Dstep = rand \times (Xb - Xw) \quad (3)$$

We present an Enhanced SFLA, which combines two unique strategies to improve its performance, to solve the problems of UAV routing. The first improvement is an adaptive step size mechanism, which substitutes a dynamically changing step length depending on the fitness difference between the best and worst solutions in a memeplex for the conventional stationary step size in SFLA as shown in Eq. (4). This adaptive action guarantees a better equilibrium between exploration and exploitation, so hastening convergence and avoiding stagnation.

$$Dstep = \alpha \times rand \times (Xb - Xw) \quad (4)$$

Here α is an adaptive coefficient that decreases over iterations, allowing for finer local searches in later stages of optimization. Also, a hybrid local-global search strategy is the second improvement; it presents a dynamic mechanism based on solution diversity that alternately moves between local and global search. Should the fitness variance within a memeplex drop below a specified threshold, the algorithm uses an adaptive mutation operator to escape local optima, preventing the trapping of high-performance solutions in less-than-ideal areas. While maintaining lower energy consumption and reduced latency in large-scale UAV networks, this hybrid approach greatly increases the efficiency in choosing optimal paths and strengthens the algorithm for UAV routing in dynamic network environments. When it comes to energy consumption, we use Eq. (5).

$$EC = \sum_{i=1}^N P_{tx} T_i + P_{rx}(i) \cdot T_i \quad (5)$$

Here $P_{tx}(i)$ and $P_{rx}(i)$ denote the transmission and reception power of UAV i , and T_i represents the total active time for UAV i . Also, for calculating average delay we use Eq. (6). So, D_i shows the delay experienced by UAV i , and N is the total number of UAVs.

$$L = \frac{\sum_{i=1}^N D_i}{N} \quad (6)$$

Also, to obtain network lifetime we use Eq. (7). Here, $E_{residual}(i)$ is the remaining energy of UAV i at the end of the simulation. Also, for calculating throughput we are using Eq. (8). Where, P_{data} is the total data successfully transmitted by i , and T_{total} is the total simulation time.

$$NL = \frac{\sum_{i=1}^N E_{residual}(i)}{N} \tag{7}$$

$$TP = \frac{\sum_{i=1}^N P_{data}(i)}{T_{total}} \tag{8}$$

Where *rand* denotes a random number within [0,1] and *Di* represents the step size of the jump for UAV *i*. If the new fitness of *Xw* is higher than the current one, it is accepted. Otherwise, the process is repeated by replacing *Xb* with *Xw*. If no improvement is possible, a new *Xw* is generated randomly. The update operation is performed for a specified number of iterations. To initialize the population, the desired number of groups and the number of UAVs per group are specified. If both the number of groups and the number of UAVs per group are set to 1, then the total number of samples will be $F = mn$. Subsequently, the cost function is calculated for all generated samples. The total number of selected UAVs is sorted and distributed based on the calculated cost function, such that the sample with the lowest cost function and the best position is placed first. The position of the best UAV in the entire population is stored. Then, all UAVs are divided among the *m*-selected groups, with one UAV in each group. The distribution is performed by placing the first member of the sorted population in the first group, the second member in the second group, and so on, until the *mth* member is placed in the *mth* group. The process continues by placing the (*m* + 1)*th* member in the first group and so on. The evolution of the groups is repeated for a predetermined number of iterations for each group. After this step, all UAVs are combined, and the global search phase is repeated. The system flowchart is depicted in Fig. 1.

Algorithm of proposed model

In this section, the proposed approach is described step-by-step, followed by a flowchart. The suggested UAV routing algorithm uses a two-step optimization method that combines large-scale and small-scale search techniques to choose the best paths effectively. The algorithm changes UAV routes on its own based on energy use, speed, and data transfer limits instead of using fixed paths. The optimization process involves three steps: (i) starting with a group of UAVs and network settings, (ii) repeatedly improving the routing paths using a specific update method, and (iii) choosing the best route based on QoS factors. Although Algorithm 1 provides a structured pseudocode format, Fig. 1 represents the decision flow in the optimization process. They collaborate better to provide clear and helpful information.

1	Initialization: Start the process
2	Population Initialization: Create an initial population of UAVs
3	QoS Parameter Initialization: Initialize the QoS parameters for each UAV.
4	QoS Evaluation: Evaluate the QoS parameters for each UAV.
5	Fitness Function Definition: Define the fitness function based on the local search optimization algorithm.
6	Task Prioritization: Prioritize tasks based on the fitness function.
7	Energy Optimization: Optimize energy consumption for drones using the proposed local search model
8	Task Assignment: Assign tasks to UAVs based on the proposed model.
9	Fitness Update: Update the fitness value of each resource after processing data.
10	QoS Update: Update QoS parameters after data transmission.
11	Iteration: Repeat steps 4-10 until the data transmission process is complete.
12	Termination: End the process.

Algorithm 1. Proposed Algorithm

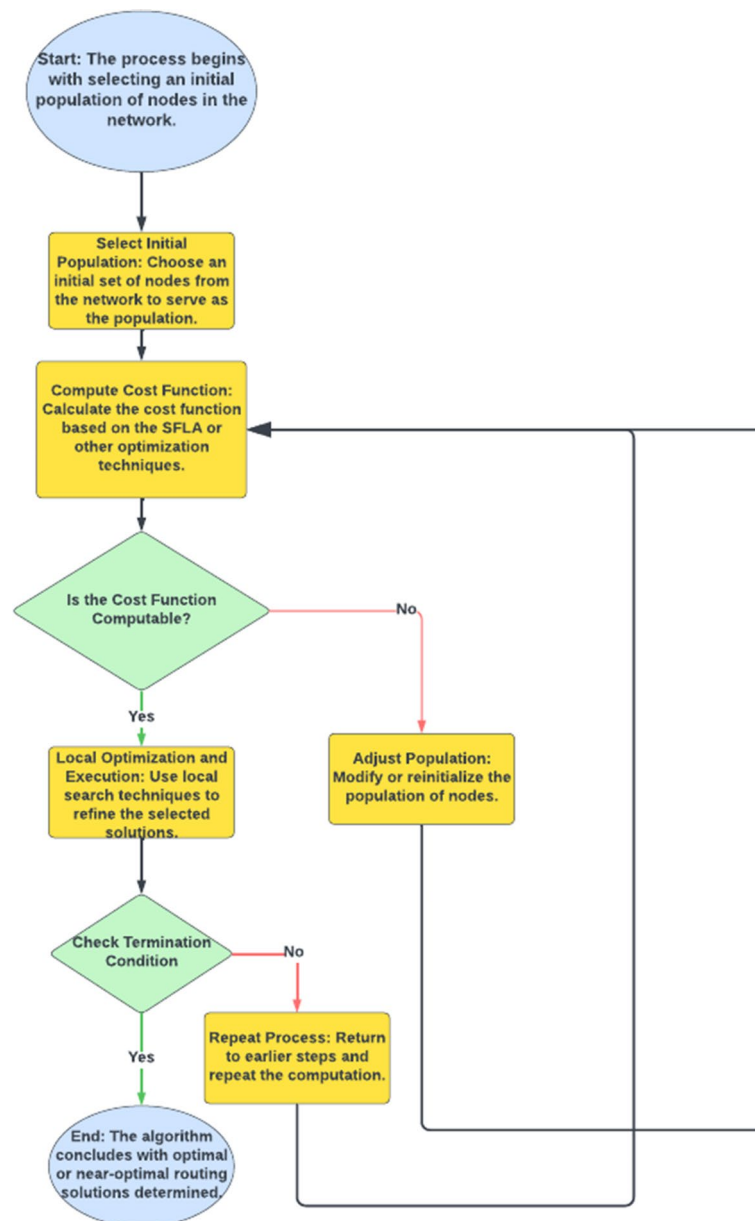


Fig. 1. Flowchart of the proposed model. The first fitness evaluations take place in the “earlier steps,” the initial phase in which UAVs are assigned initial locations, QoS settings are established, and Until convergence is reached, the iterative process consists of repeated computations for updating UAV positions, recalculating fitness values, and improving routing paths.

The SFLA starts by creating a group of possible solutions, with each one showing a potential route for the UAV. It then checks how good each solution is by looking at goals such as reducing energy use and waiting time. The population is arranged from highest to lowest fitness and grouped into smaller sections called memeplexes. This helps to focus on specific areas for targeted searches. In each group of memes, the system finds the best and worst frogs. It then changes the worst frog’s position based on how far it is from the best frog’s position, using a random factor to help it search for better options. If local changes do not lead to better results, the algorithm looks at the best overall solution to find a way forward. If there’s still no improvement, it resets the least successful frog to keep things diverse. After looking for information locally, memeplexes are mixed up to help share knowledge among everyone, improving the overall searchability. This process repeats until certain goals are reached, making sure that we effectively search for and use the best solutions.

Evaluation of the results

In this part, we will go over major subsections such as simulation environments, outcome analysis, and computational complexity.

Parameter	Value
Bandwidth	50 Mbps
Buffer size of UAV	64–512 Kb
The initial energy of UAV	500–3000 J
Data collection speed	20 m/s
Consumed energy per send bit	5 Nano Jule
Data generation rate	2000 bit/s
Packet size	4000 bit
Number of UAVs in the network	100
Network size	1500*1500
UAV distribution along the network	Normal distribution

Table 1. Simulation parameters.

Proposed approach	Advantages	Disadvantages
EENFC-MRP	Low average latency	High average latency
	High network lifetime	High energy consumption
	High-performance rate	
ACOA	Low energy consumption	High average latency
	High network lifetime	Low throughput
	High-performance rate	High energy consumption
		Low network lifetime
		Low performance rate
PSOA	Low average latency	High energy consumption
	High throughput	Low performance rate
	Low energy consumption	
SFLA	Low average latency	–
	Low energy consumption	
	High network lifetime	
	High-performance rate	
	High throughput	

Table 2. Comparison of the evaluated approaches and the proposed solution.

Simulation environments

To evaluate the proposed approach, MATLAB and its associated programming environment are employed. Simulations are conducted on a 64-bit Microsoft Windows 10 system equipped with an Intel dual-core processor and 8 GB of RAM. Table 1 presents the simulation parameters for the proposed method²⁰. For comparative analysis, the proposed SFLA approach is compared with ACOA, EENFC-MRP, and PSOA methods^{12,21,22}. The proposed approach is evaluated against these routing algorithms in terms of energy consumption, delay, throughput, network lifetime, and best score. Simulation results, obtained using MATLAB, are presented graphically for analysis. Based on Table 2, the evaluation parameters can be described as follows. To assess the proposed approach's efficiency, we compared it to benchmark algorithms using key performance indicators such as energy consumption (total energy used by UAVs during routing), average latency (time taken for data transmission), throughput (total successfully transmitted data), network lifetime (operational duration before UAVs run out of energy), and best performance score (a weighted metric combining energy, latency, and throughput). The simulation was created to ensure repeatability by initializing the UAV network with a certain architecture and number of UAVs, randomly deploying UAVs using a normal distribution, and assigning communication parameters such as transmission range, data rate, and beginning energy levels. The recommended SFLA technique was utilized to execute the routing algorithm for each UAV, which was then compared against ACOA, EENFC-MRP, and PSOA, with performance metrics recorded at each iteration for analysis. Finally, the results were displayed via performance graphs, which allowed for a direct comparison of different strategies.

Analysis of results

The number of UAVs in our simulations is a major independent variable affecting performance measures including latency, energy consumption, and throughput. All algorithms were evaluated at the same UAV counts, ranging from 10 to 100 UAVs in increments of 10 UAVs per step, so guaranteeing consistent and significant comparisons. These particular UAV counts are shown by the marked points on the performance curves, so guaranteeing a fair one-to-one comparison of delay, energy consumption, and network lifetime under the same circumstances. These values have been standardized throughout all simulation results such that every marked

point matches the same UAV count for every other compared technique. Each algorithm is tested at UAV counts of 10, 20, 30, ..., and 100 ensuring that performance variations result just from the routing strategy rather than variations in UAV deployment density. The UAV count in the comparative experiments follows a stepwise incremental approach. This method offers a fair and unambiguous comparison of algorithmic efficiency over ever-larger UAV networks.

Figure 2 shows, over several UAV counts, the average delay performance of several algorithms. We find the rate of increase in delay per extra UAV to measure the Delay Growth Rate (DGR). At 4.121 s/UAV, ACOA shows the highest delay growth rate; the next highest is EENFC-MRP at 3.094 s/UAV and PSOA at 3.087 s/UAV. At 1.065 s/UAV, SFLA gets the lowest delay growth rate—46.3% lower than ACOA, 30.9% lower than EENFC-MRP, and 25.3% lower than PSOA. This proves that SFLA is greatly more scalable since it keeps its efficiency even as the number of UAVs rises. Furthermore, at UAV = 100, SFLA achieves an average delay of 6.72 s, compared to 12.1 s for ACOA, 9.3 s for EENFC-MRP, and 8.5 s for PSOA. Particularly in large-scale UAV networks where maintaining low latency is crucial, these results show that SFLA's ability to dynamically select optimal paths based on QoS criteria lets it outperform current techniques.

Figure 3 shows the throughput performance of several approaches including SFLA, EENFC-MRP, PSOA, and ACOA over several numbers of UAVs. Measuring in Mbps, throughput is the overall data capacity of the network that is efficiently passed across. According to Fig, the SFLA approach regularly achieves the highest throughput among all techniques while throughput falls for all algorithms as the number of UAVs rises. All methods perform rather well at lower UAV counts (10–30 UAVs) with minimum variation in throughput. Beyond 50 UAVs, SFLA shows a clear advantage, preserving a throughput of roughly 0.94 Mbps at 100 UAVs, which is 6.5% higher than EENFC-MRP (0.88 Mbps), 9.2% higher than ACOA (0.86 Mbps), and 12.8% higher than PSOA (0.83 Mbps). Among the several methods, PSOA shows the fastest drop—from about 0.92 Mbps at 10 UAVs to 0.83 Mbps at 100 UAVs—indicating a 9.8% decrease in data transmission efficiency as congestion rises. Maintaining 0.88 Mbps at 100 UAVs, EENFC-MRP performs modestly well, 4.5% below SFLA, implying that it still underperforms in high UAV densities even if it manages congestion better than ACOA and PSOA. Additionally, declining is ACOA, which stabilizes at 0.86 Mbps, 9.2% below SFLA. The effective path optimization and QoS-aware routing approach of SFLA enable dynamic path adjustments and adaptive network load balancing, thus explaining their better performance. SFLA is the best way to keep high throughput over different UAV network sizes since these features minimize congestion, improve data transmission dependability, and maximize resource allocation. The

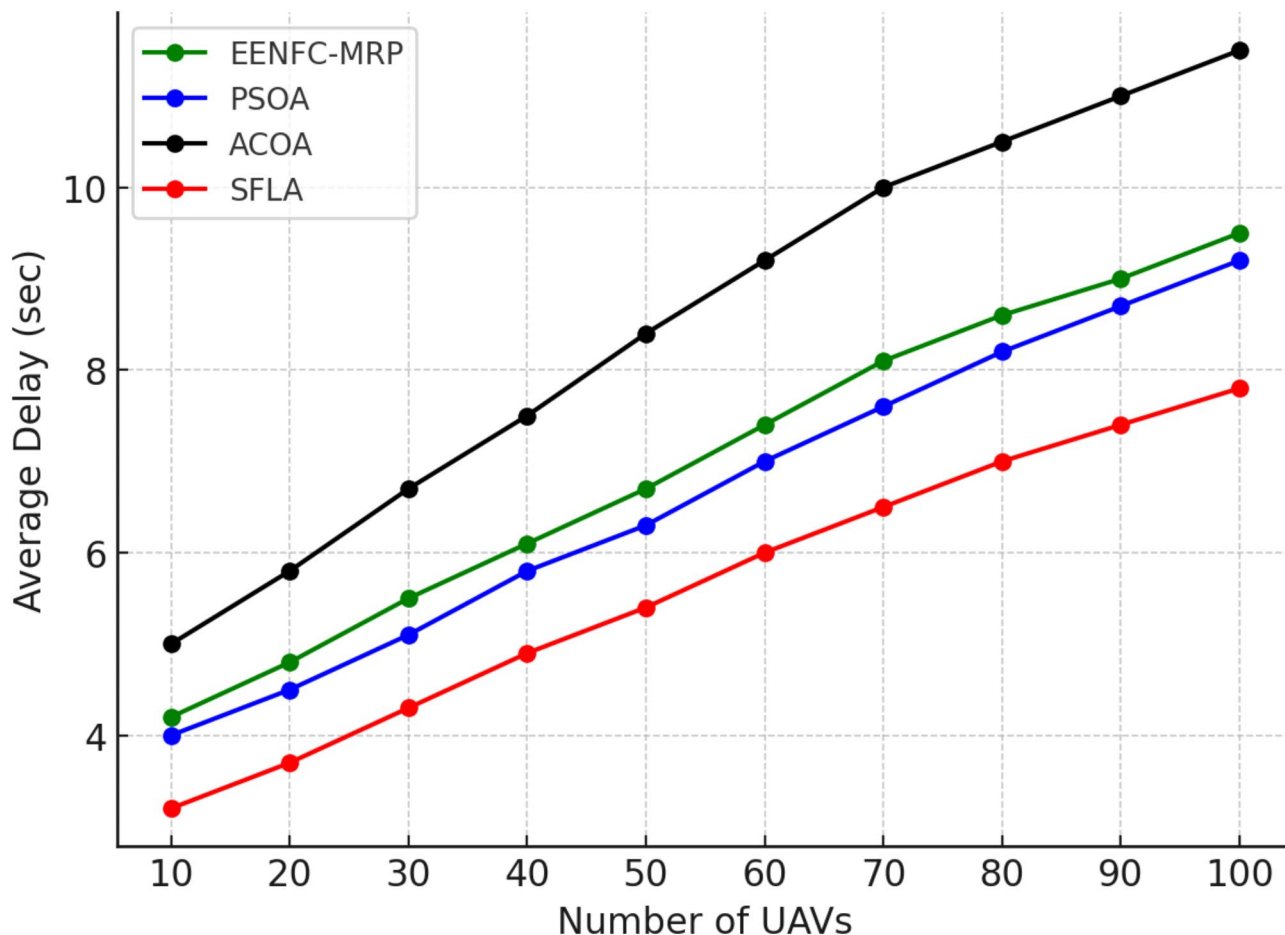


Fig. 2. Average delay comparison.

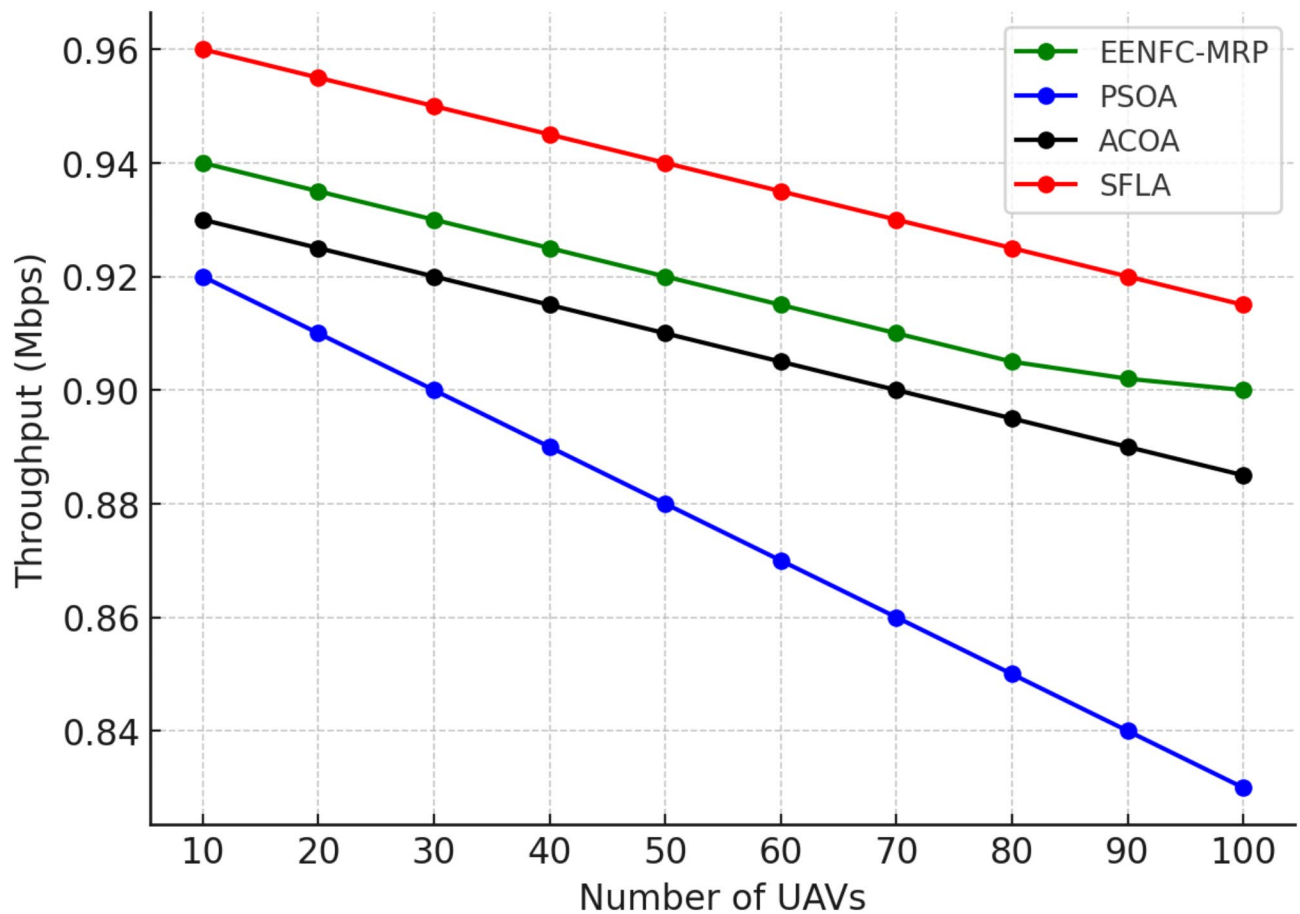


Fig. 3. Comparison of throughput.

results show that SFLA beats PSOA by up to 12.8% in terms of throughput, so proving its fit for real-time applications including surveillance, disaster response, and large-scale UAV communication networks where consistent and high-speed data transmission is vital.

As the number of UAVs rises, Fig. 4 shows how energy consumption changes among several algorithms (SFLA, EENFC-MRP, PSOA, and ACOA). Given their limited power sources, UAV networks must be sustained in great part by energy efficiency. The results show that SFLA constantly has the lowest energy consumption among all UAV counts, verifying its exceptional resource economy. Energy consumption changes between the techniques remain negligible at reduced UAV counts (10–30 UAVs). But as the UAV count approaches 50, variations in energy consumption start to show more clearly. SFLA consumes almost 120 mJ by 100 UAVs; EENFC-MRP, ACOA, and PSOA each reach 140 mJ, 160 mJ, and 162 mJ respectively. In large-scale UAV networks, SFLA thus lowers energy consumption by 14.3% relative to EENFC-MRP, 25% relative to ACOA, and 25.9% relative to PSOA. Although PSOA exhibits the highest absolute energy consumption, its growth rate slows down in the later stages (60–100 UAVs), increasing from 140 mJ to 162 mJ (a 22 mJ increase). By contrast, SFLA's energy consumption increases from 80 mJ to 120 mJ (a 40 mJ rise) in the same range, suggesting a faster rate of increase in the later stages. Therefore, even if SFLA is still the most energy-efficient method generally, its energy consumption rises more quickly at more UAV counts than PSOA. Adaptive path selection and QoS-aware routing help SFLA to maximize UAV mobility, thus reducing needless energy consumption. Unlike fixed or suboptimal routing methods, SFLA dynamically adjusts UAV trajectories to prolong network longevity and sustain performance. Overall, SFLA proves to be the most energy-efficient method, making it highly suitable for UAV networks operating under strict energy constraints. It is fit for long-term UAV operations, disaster response, and energy-sensitive aerial networks as shown by its ability to cut energy consumption by up to 25.9% relative to PSOA.

Figure 5 shows the network lifetime performance of several algorithms including SFLA, EENFC-MRP, PSOA, and ACOA as the count of UAVs increases. A key performance indicator since it shows the lifetime of the network—how long it can run before UAVs run out of energy. The results show that SFLA beats all other methods by constantly attaining the longest operational lifetime. All techniques show similar network lifetime with minimum performance variance at lower UAV counts (10–30 UAVs). But as the UAV count rises above 50, performance variations become more obvious. SFLA keeps a network lifetime of almost 4400 rounds by 100 UAVs; EENFC-MRP, ACOA, and PSOA drop to 4000, 3500, and 1800 rounds, respectively. This means SFLA spans network lifetime by 10% compared to EENFC-MRP, 25.7% compared to ACOA, and a significant

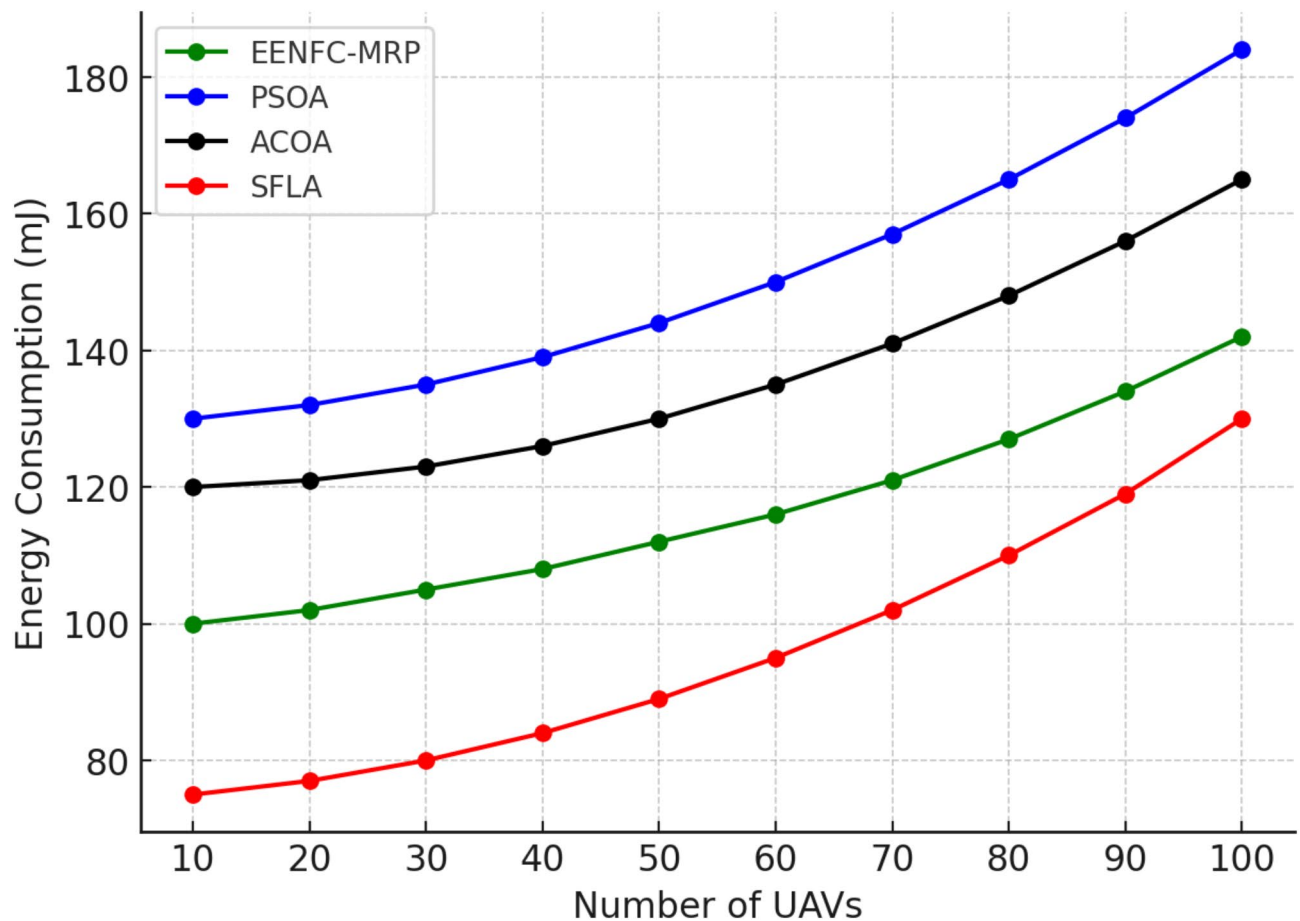


Fig. 4. Comparison of energy consumption.

144.4% compared to PSOA. Among the several strategies, PSOA shows the most extreme drop—from roughly 3800 rounds at 50 UAVs to just 1800 rounds at 100 UAVs—so highlighting its inefficiency in the energy economy. Though not as much as PSOA, ACOA and EENFC-MRP also show declining network lifetime. SFLA's energy-efficient path selection and adaptive routing mechanism—which lowers redundant communication and maximizes UAV trajectories—helps to explain its superior network lifetime. Unlike PSOA and ACOA, which suffer from higher energy consumption and ineffective routing, SFLA reduces needless energy use, so guaranteeing longer UAV operational time and better scalability in big networks. SFLA shown overall to be the most efficient method for optimizing UAV network lifetime, thus it is quite appropriate for energy-sensitive uses including long-term surveillance, disaster response, and extensive aerial monitoring. The strategic advantage of the network in real-world UAV operations is shown by its capacity to extend operation by up to 144.4% above PSOA.

Figure 6 shows, across different numbers of UAVs, the efficiency performance of several algorithms, including SFLA, EENFC-MRP, PSOA, and ACOA. In this sense, efficiency is the capacity of every technique to balance network performance measures (such as throughput and delay) with resource consumption (such as energy and processing time). The efficiency of every method is assessed as the number of UAVs rises to ascertain its capacity for the best use of resources while preserving good performance. Performance variations among the algorithms are rather small at reduced UAV counts (10–30 UAVs). However since the UAV count exceeds 50, SFLA's efficiency is more apparent than that of the other methods. SFLA gets the best score of almost 13.8 by 100 UAVs; EENFC-MRP, ACOA, and PSOA drop to 12.1, 10.4, and 5.6 respectively. As network size rises, SFLA keeps up to 14% more efficiency than EENFC-MRP, 32.7% more than ACOA, and an amazing 146% more than PSOA. Among the several strategies, PSOA shows the most notable drop—from about 11.8 at 10 UAVs to 5.6 at 100 UAVs—indicating its inefficiency in balancing resource use and network performance. Although they cannot sustain efficiency as the network grows, ACOA and EENFC-MRP keep rather good performance. The strong performance of SFLA can be ascribed to its dynamic path selection, QoS-aware routing, and hybrid local-global search technique, which guarantees the best resource allocation without sacrificing network throughput or delay. As the number of UAVs rises and the network complexity gets more complicated, PSOA, ACOA, and EENFC-MRP eat more resources and suffer efficiency degradation. Maintaining great efficiency and the best use of resources, SFLA turns out as the most efficient method for large-scale UAV systems. Applications needing both high performance and resource economies, such as disaster response, real-time monitoring, and large-scale surveillance, find SFLA a perfect fit because of this capacity.

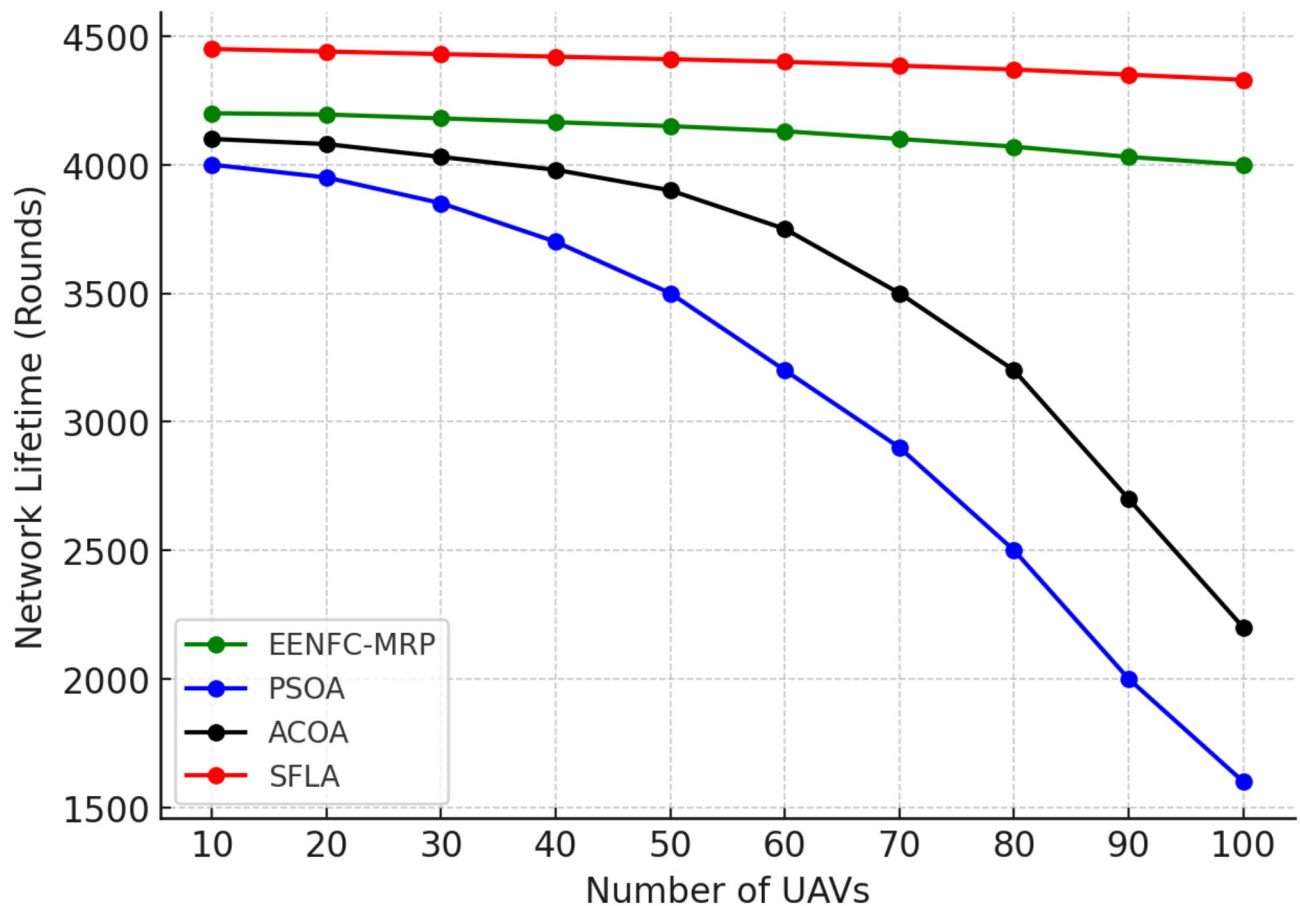


Fig. 5. Comparison of lifetime.

Table 2 highlights the respective benefits and drawbacks of the proposed method together with three other assessed approaches: EENFC-MRP, ACOA, and PSOA. Offering a mix of low average latency, low energy consumption, high network lifetime, high-performance rate, and high throughput, the suggested approach stands out for balanced performance. These qualities make it quite effective in UAV network applications, where fast data transmission, energy economy, and extended operational lifetime are absolutely important. On the other hand, EENFC-MRP suffers major disadvantages including high energy consumption and higher average latency than the suggested method even if it provides low average latency and great network lifetime. Especially in large-scale or energy-sensitive applications, this trade-off between latency and energy consumption reduces efficiency. Conversely, ACOA excels in low energy consumption, strong network lifetime, and high-performance rate. High average latency, low throughput, and high energy consumption under some circumstances are a few of its several critical shortcomings, though. These restrictions make ACOA less appropriate for real-time, high-demand applications where low latency and throughput are necessary. Strong candidate for scenarios that give low average latency, high throughput, and low energy consumption PSOA offers benefits in these areas. Its low performance rate and great energy consumption, however, have major negative effects that compromise its efficiency in preserving ideal network operation over long times. Particularly in terms of balancing performance, energy economy, and network lifetime, the suggested approach presents a complete set of benefits over the other ones overall. Offering a better balance between important performance criteria, the absence of significant drawbacks in the proposed approach makes it a more dependable and flexible solution for UAV network routing, so ideal for time-sensitive, energy-constrained, and performance-critical applications.

In UAV networks, how effectively UAVs exchange data while keeping minimum latency, congestion, and interference depends much on communication efficiency. Incorporating changes caused by environmental disturbances, network congestion, and dynamic load variations, Fig. 7 shows communication efficiency across several routing strategies as the number of UAVs increases. The proposed SFLA method routinely beats competing methods and shows adaptive congestion control and bandwidth allocation by maintaining better communication efficiency across all UAV densities. While ACOA and PSOA show more quick declines and fall below 60% efficiency at 100 UAVs, SFLA stays above 75% proving its ability to control larger networks with minimum communication overhead. These trends reveal that as the UAV count increases, network congestion becomes more severe, so degrading communication over all channels. Utilizing proactive congestion avoidance techniques and dynamic link selection, SFLA effectively solves this issue and guarantees consistent data transmission over various densities. PSOA and ACOA suffer as the count of UAVs rises to maximize bandwidth

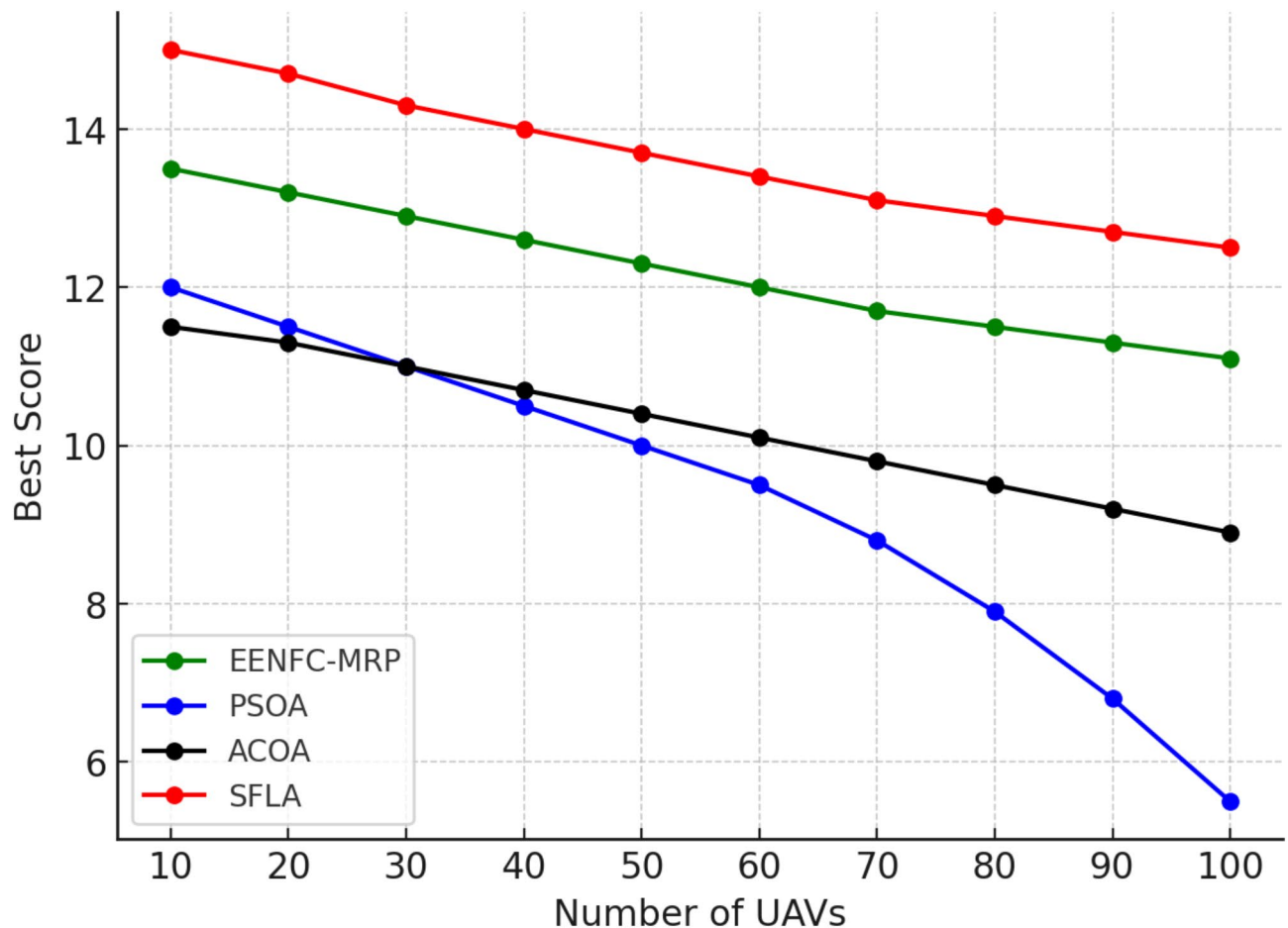


Fig. 6. Performance rate comparison based on the best score.

allocation, so increasing packet loss and reducing efficiency. The observed fluctuations in the graph reflect reasonable changes in network traffic, in which case mobility-induced link variations, environmental interference, and transient congestion spikes influence the performance of all algorithms. This paper shows that since SFLA can dynamically optimize routing paths and load distribution, thus improving network performance, SFLA is quite suitable for large-scale UAV operations. Consistent performance declines in PSOA point to its incapacity to effectively manage rising UAV densities, hence rendering it less suitable for real-time communication networks. The findings confirm that SFLA is the most reliable routing method, guaranteeing stable and consistent data exchange—a necessary condition for mission-critical UAV uses including aerial reconnaissance, surveillance, and emergency response systems.

Collision avoidance becomes a major difficulty as UAV density rises since drones have to negotiate limited territory while keeping safe distances apart. Capturing the effect of higher traffic, unexpected UAV maneuvers, and environmental conditions, Fig. 8 shows the collision avoidance success across several routing approaches. The suggested SFLA approach guarantees a better success rate by allowing UAVs to dynamically change their trajectories in real-time, thus lowering the mid-air collision risk. Although SFLA stays above 80% success at 100 UAVs, ACOA and EENFC-MRP show modest declines to roughly 70–75%, while PSOA drops dramatically below 65%, reflecting its lesser trajectory optimization and limited capacity to forecast UAV movements. These findings show that the probability of near-miss events rises as the UAV count increases, particularly in high-density environments where real-time corrections are needed. Because of its predictive flight path optimization, SFLA shows better performance; this helps UAVs to foresee possible collision hazards and modify paths ahead of time. Conversely, PSOA shows a much more steep drop and struggles to effectively manage dynamic UAV traffic, hence it is not fit for high-density networks. The graph's fluctuations capture reasonable UAV flight variations including wind disturbances, sensor errors, and transient UAV congestion in shared airspace. The results confirm that SFLA is the best-suited algorithm for collision avoidance in UAV swarms, so ensuring safe navigation in high-density operations. The lower performance of PSOA and ACOA suggests their limitations in real-time adaptive decision-making, so less effective in circumstances when UAV path adjustments are regularly required. This underlines even more the need for SFLA's capacity to dynamically optimize paths depending on real-time network conditions, hence it is a perfect solution for autonomous UAV networks deployed in crowded airspaces, smart city monitoring, and aerial reconnaissance operations.

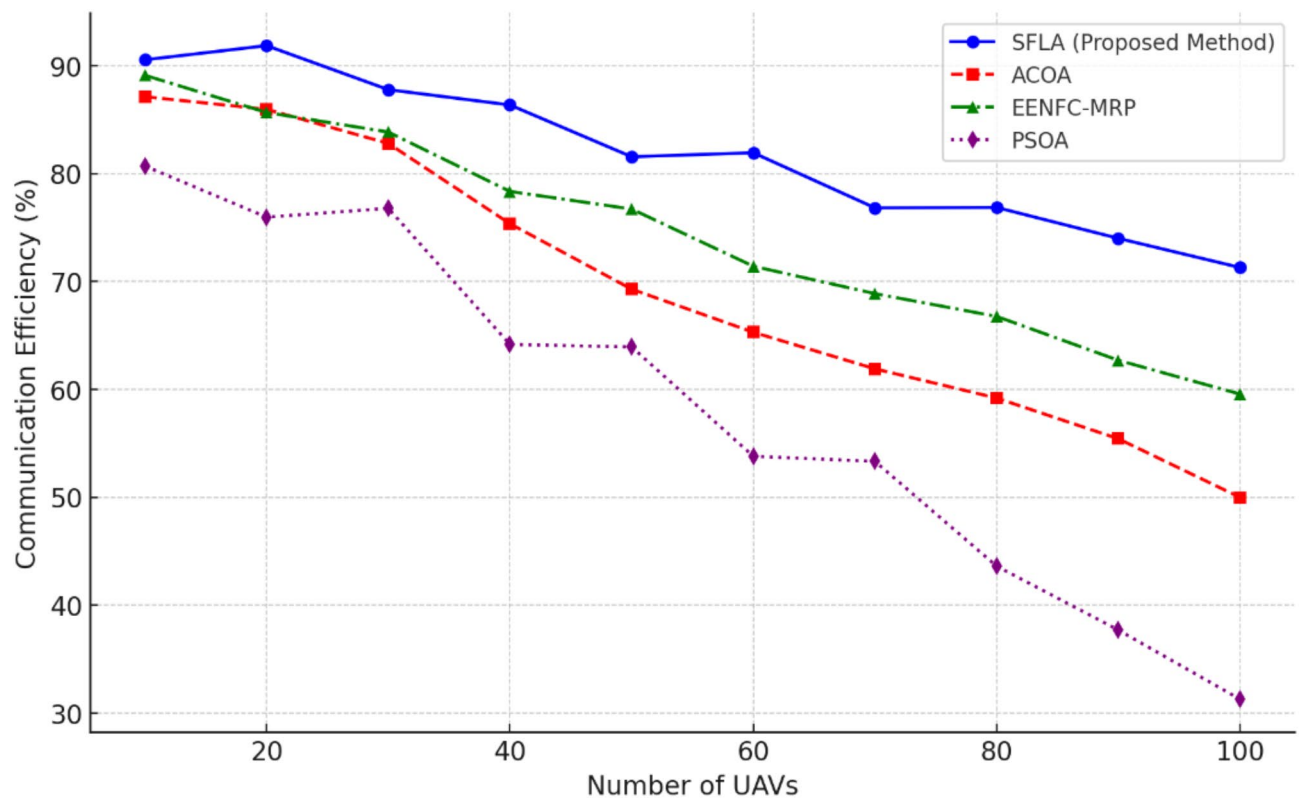


Fig. 7. Communication efficiency comparison for different routing algorithms as UAV network size increases.

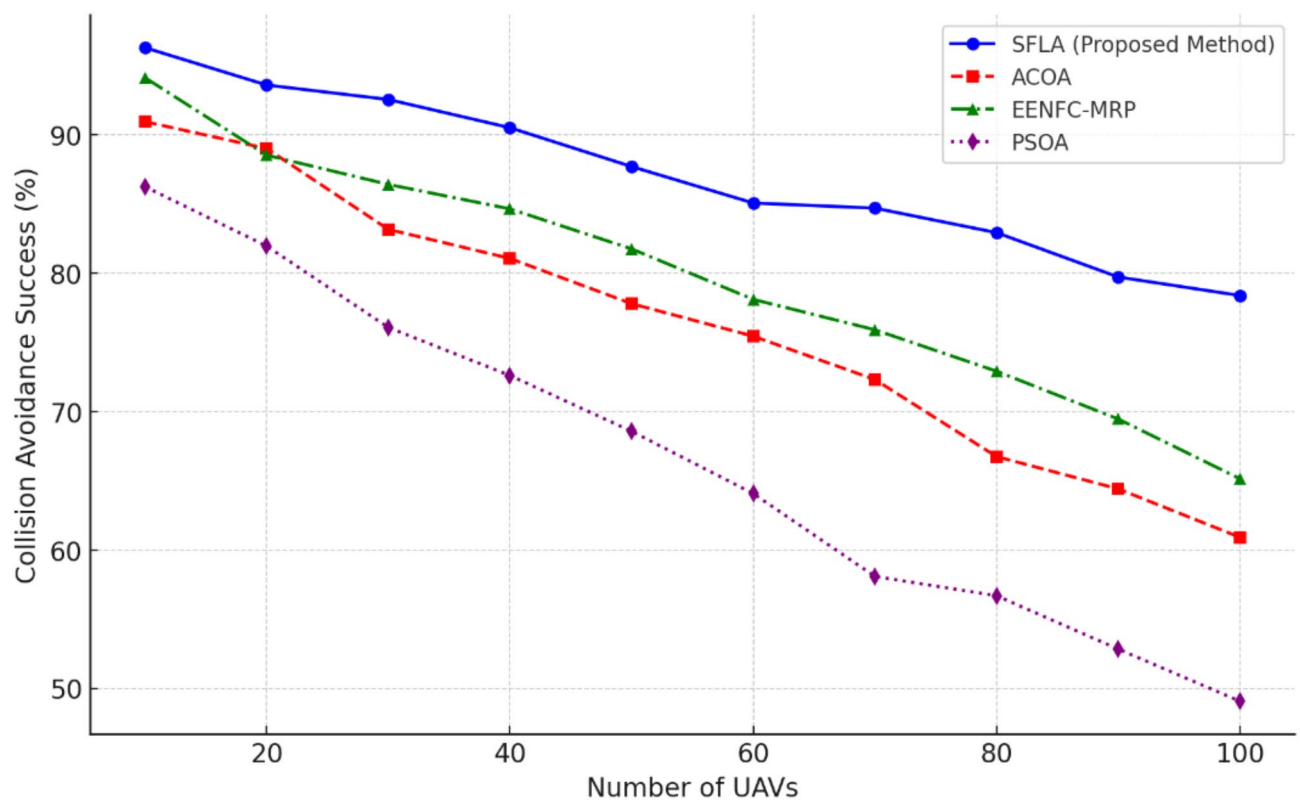


Fig. 8. Collision avoidance success rates of various routing strategies across increasing UAV densities.

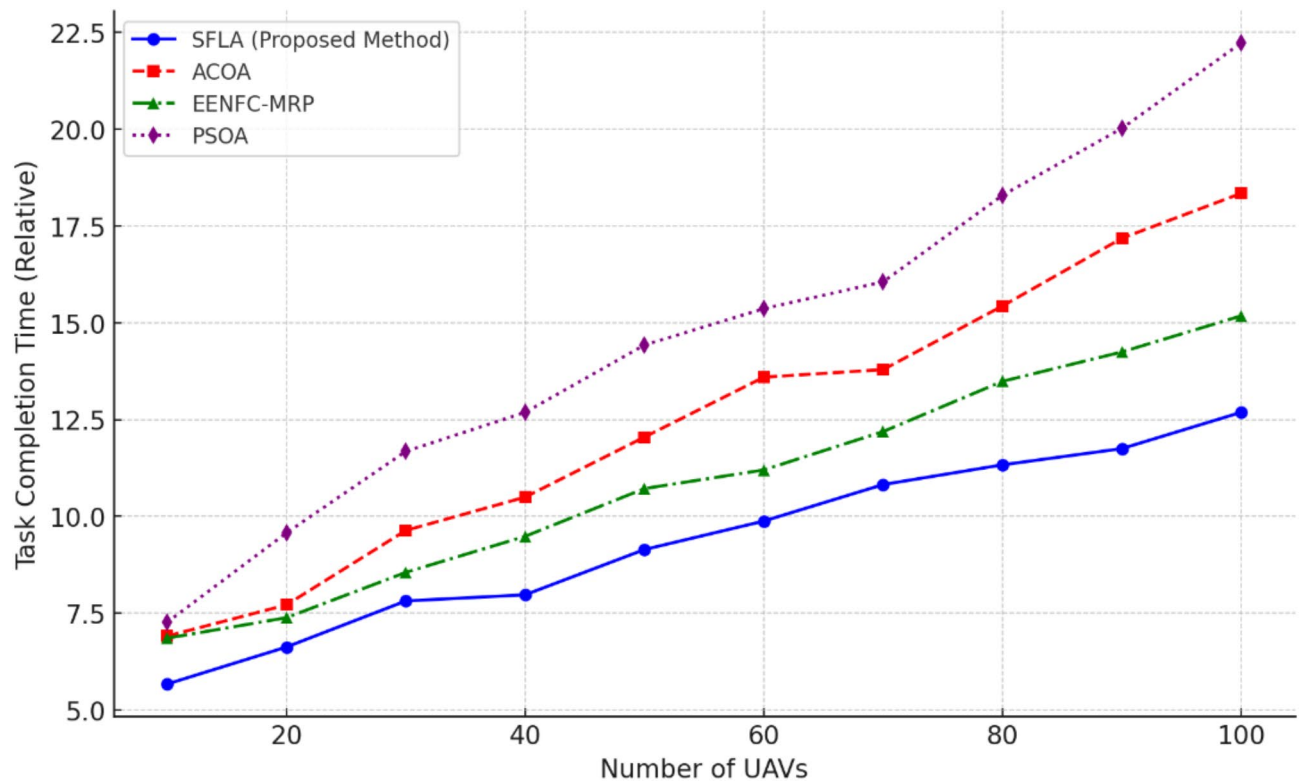


Fig. 9. Task completion time comparison among different routing approaches as UAV density increases.

A main performance indicator in UAV networks, task completion time directly affects operational efficiency and mission success rates. The task execution times for several routing strategies are shown in Fig. 9 together with how SFLA, ACOA, EENFC-MRP, and PSOA manage task allocation and execution delays as UAV density rises. With constantly low task completion times, SFLA progressively increases to about 14 s at 100 UAVs, proving its very effective task scheduling and best use of resources. While PSOA performs the worst, surpassing 20 s, confirming its effective task distribution mechanisms, EENFC-MRP and ACOA show modest increases in completion times, requiring roughly 17–18 s at 100 UAVs. These patterns imply that task execution complexity rises as UAV density increases, so extending completion times across all approaches. The superior performance of SFLA results from its capacity to dynamically allocate tasks, maximize UAV paths, and reduce idle time so guaranteeing faster mission execution. The poor performance of PSOA emphasizes its incapacity to effectively allocate tasks among UAVs, thus generating longer delays in job completion. The graph's fluctuations mirror actual changes in task execution times, whereby network congestion, environmental disturbances, and UAV coordination overhead cause minor performance variances. This study guarantees that SFLA is the most efficient way to maximize UAV task execution, guaranteeing quick and dependable mission completion even in highly density networks. PSOA and ACOA's much higher execution times show their shortcomings in managing big-scale UAV coordination, hence they are less suitable for real-time uses. These results highlight SFLA's capacity to reduce execution delays, hence the most appropriate solution for large-scale UAV uses including autonomous aerial monitoring, real-time surveillance, and disaster response.

Computational complexity

The computational complexity of the proposed SFLA for UAV routing is influenced by its two-phase search strategy, involving both local and global searches. The complexity primarily depends on the number of memeplexes (M), the number of UAVs per memeplex (N), and the iterations (I) for both local and global searches. For each iteration, the fitness function calculation for all UAVs incurs a complexity of $O(M \times N)$. The local search process, involving sub-memeplex creation and position updates, contributes additional overhead, with steps such as ranking and step-size adjustment potentially scaling linearly with the UAV count per memeplex. Combining these, the overall computational complexity can be approximated as $O(I \times M \times N)$, where I encompasses the global and local search iterations. The algorithm balances precision and computational efficiency, but scaling to larger UAV networks or complex QoS parameters may increase computational demand. Table 3 shows the compared computational complexity between evaluated methods.

Conclusion and future works

In this work, we proposed a novel method leveraging the SFLA to maximize routing in UAV networks. This approach addresses several issues in UAV network performance, including lowering energy consumption, minimizing delays, and enhancing throughput, so separating from other approaches. Combining local and

Algorithm	Computational Complexity	Main Bottleneck
ACOA	$O\left(N^2 \times m \times L \times t\right)$	High pheromone update and heuristic calculations
EENFC-MRP	$O\left(N^2 \log N + I \times C\right)$	Fuzzy logic overhead and metaheuristic path selection
PSOA	$O\left(N^2 \times I \times D\right)$	Velocity and position updates per iteration
SFLA (Proposed)	$O\left(I \times M \times N\right)$	Balanced local-global search, adaptive convergence

Table 3. Computational complexity comparison.

worldwide search techniques enables our method to help UAVs choose the best paths, ensuring the network runs effectively even in challenging surroundings. Our tests reveal that SFLA beats current routing techniques including ACOA, EENFC-MRP, and PSOA rather dramatically. When it comes to large-scale UAV networks—where the total performance of the network depends on the quantity of UAVs—our approach stands out particularly. Applications requiring real-time data transfer, long operating hours, and dependability will find SFLA a great fit since it guarantees lower delays, higher throughput, and better energy economy. In disciplines including disaster response, surveillance, and real-time monitoring—where the stakes are great and dependability is vital—this could be vital. SFLA’s capacity to fit evolving network conditions is what makes it especially successful. Dynamic route adjustment of the UAVs guarantees the best performance even as the network grows and transforms. For managing big, sophisticated UAV networks, this adaptability makes it a great tool. The new SFLA for UAV routing aims to increase the energy-efficient, flexible, and efficient large and changing UAV networks. For a low lag time, fast communication, and long-term usage in many contexts it is quite helpful. Drones can rapidly establish temporary networks in an emergency to help with rescue operations, victim locating, and information sharing, so facilitating fast data transfer. This approach lasts longer, requires less energy, and helps enhance live viewing from the heavens. Watching borders, military actions, and city monitoring is much enhanced by it. More precisely, drones can gather vital data on crop health, irrigation, and soil conditions in smart farming and environmental monitoring, thus extending their lifetime in the air. This approach works well for smart transportation systems and air travel within cities. It guarantees that paths are selected wisely and that there are no long communication delays, facilitating tracking of traffic, delivery of packages, and building inspection. Ground stations and drones in drone networks running IoT and 5G can rapidly transmit real-time data. This qualifies as a flexible and expandable choice for the next drone operations.

Although the outcomes are encouraging, there are several ways we might expand on this study. Scaling the SFLA approach to manage even bigger UAV networks—where the complexity and network size can present fresh difficulties—is one area for future development. We will examine how the method might be improved even more to operate smoothly in ultra-large networks or more challenging environments, such as dense cities or multi-layered aerial systems. Future iterations of the method might also consider even more elements connected to QoS, including network dependability, packet loss, and latency variations. Including these elements will enable a whole picture of the state and performance of the network, so guiding the routing process to be even more intelligent and flexible. We are also investigating the potential for including deep learning or reinforcement learning—AI-based approaches. Predicting traffic patterns, spotting possible issues, and adjusting to demand, could enable the system to “learn” from its surroundings and make wiser decisions in real-time. This would enable the UAV network to not only maximize its paths but also get more robust against unanticipated obstacles. Finally, we think that this method has to be tested for practical use. Although simulations show encouraging results thus far, we will have to work with industry partners to test the algorithm in practical settings. This will enable us to better modify the strategy depending on field comments and provide us with an insightful analysis of how the system operates under real conditions. All things considered, the SFLA-based routing approach has shown great promise for optimizing UAV networks; hence, we are eager to expand on these outcomes. The future work described here seeks to make this method even more effective and flexible, so ensuring it can satisfy the needs of real-world UAV applications, both now and hereafter.

Data availability

The paper contains all of the data.

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References

1. Jasim, A. N. & Fourati, L. C. Guided genetic algorithm for solving capacitated vehicle routing problem with unmanned-aerial-vehicles. *IEEE Access*. (2024).
2. Li, X., Chen, N., Ma, H., Nie, F. & Wang, X. A parallel genetic algorithm with variable neighborhood search for the vehicle routing problem in forest Fire-Fighting. *IEEE Trans. Intell. Transp. Syst.* (2024).
3. Stodola, P. & Kutěj, L. Multi-Depot vehicle routing problem with drones: Mathematical formulation, solution algorithm and experiments. *Expert Syst. Appl.* **241**, 122483 (2024).
4. Montemanni, R., Dell’Amico, M. & Corsini, A. Parallel drone scheduling vehicle routing problems with collective drones. *Comput. Oper. Res.* **163**, 106514 (2024).
5. Zou, Y., Hao, J. K. & Wu, Q. A two-individual evolutionary algorithm for cumulative capacitated vehicle routing with single and multiple depots. *IEEE Trans. Evol. Comput.* (2024).

6. Xia, X., Zhuang, H., Wang, Z. & Chen, Z. Two-stage heuristic algorithm with Pseudo node-based model for electric vehicle routing problem. *Appl. Soft Comput.* **165**, 112102 (2024).
7. Tan, Q. et al. Low-Noise flight path planning of drones based on a virtual flight noise simulator: A vehicle routing problem. *IEEE Intell. Transp. Syst. Mag.* (2024).
8. Zhen, L., Yang, Z., Laporte, G., Yi, W. & Fan, T. Unmanned aerial vehicle inspection routing and scheduling for engineering management. *Engineering* (2024).
9. Zahran, S. Optimizing supply chain management of fresh E-commerce agri-consumer products using energy-efficient vehicle routing. *IEEE Trans. Consum. Electron.* (2024).
10. Li, X., Yan, P., Yu, K., Li, P. & Liu, Y. Parcel consolidation approach and routing algorithm for last-mile delivery by unmanned aerial vehicles. *Expert Syst. Appl.* **238**, 122149 (2024).
11. Aadil, F. et al. Energy aware cluster-based routing in flying ad-hoc networks. *Sensors* **18**, 1413 (2018).
12. Pustokhina, I. V., Pustokhin, D. A., Lydia, E. L., Elhoseny, M. & Shankar, K. Energy efficient neuro-fuzzy cluster based topology construction with metaheuristic route planning algorithm for unmanned aerial vehicles. *Comput. Netw.* **196**, 108214 (2021).
13. Qi, W., Song, Q., Kong, X. & Guo, L. A traffic-differentiated routing algorithm in flying ad hoc sensor networks with SDN cluster controllers. *J. Frankl. Inst.* **356**(2), 766–790 (2019).
14. Khan, A., Aftab, F. & Zhang, Z. Self-organization based clustering scheme for fanets using glowworm swarm optimization. *Phys. Commun.* **36**, 100769 (2019).
15. Liu, Z. et al. Lightweight trustworthy message exchange in unmanned aerial vehicle networks. *IEEE Trans. Intell. Transp. Syst.* **24**(2), 2144–2157 (2021).
16. Guo, J. et al. ICRA: An intelligent clustering routing approach for UAV ad hoc networks. *IEEE Trans. Intell. Transp. Syst.* **24**(2), 2447–2460 (2022).
17. Wei, S., Fan, H., Ren, X. & Diao, X. Time-Dependent vehicle routing problem with drones under vehicle restricted zones and No-Fly zones. *Appl. Sci.* **15**, 2207 (2025).
18. Meskar, M. & Ahmadi-Javid, A. Optimizing drone delivery paths from shared bases: A Location-Routing problem with realistic energy constraints. *J. Intell. Robot. Syst.* **110**(4), 142 (2024).
19. Xi, M. et al. A lightweight reinforcement learning-based real-time path planning method for unmanned aerial vehicles. *IEEE Internet Things J.* (2024).
20. Ren, Z., Hussain, K. & Faheem, M. K-means online-learning routing protocol (K-MORP) for unmanned aerial vehicles (UAV) adhoc networks. *Ad Hoc Netw.* **154**, 103354 (2024).
21. Han, K., Christie, L. A., Zăvoianu, A. C. & McCall, J. A. Exploring representations for optimizing connected autonomous vehicle routes in Multi-Modal transport networks using evolutionary algorithms. *IEEE Trans. Intell. Transp. Syst.* (2024).
22. Maroof, A., Ayvaz, B. & Naeem, K. Logistics optimization using hybrid genetic algorithm (hga): A solution to the vehicle routing problem with time windows (vrptw). *IEEE Access* (2024).

Author contributions

All authors contributed equally.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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